Benchmark: **MA.7.AR.4.2** · Constant of proportionality Student code: _____ Calculator: yes

Learning goal: I can find the constant of proportionality. (circle your level after scoring)

4	I can explain what the constant of proportionality means in the situation.
3	I can find the constant of proportionality from a table, graph, or written description.
2	I can find a unit rate from a ratio.
1	I need help to start this benchmark.

1. In a proportional relationship, 64.05 cups of water are used in 7 minutes.

What is the constant of proportionality, in cups of water per minute?

A. 9.15

B. 64.05

C. 71.05

D. 0.11

2. The daily fee for docking a boat at a marina is proportional to the length of the boat. The table shows the fee for four different boat lengths.

Boat length (feet)	Daily fee
20	$62.00
22	$68.20
24	$74.40
28	$86.80

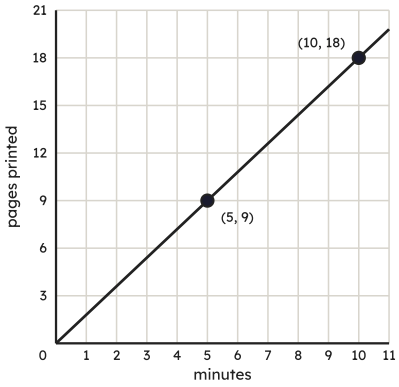
What is the constant of proportionality, in dollars per foot?

Turn over →

Benchmark: **MA.7.AR.4.2** · Constant of proportionality Student code: _____

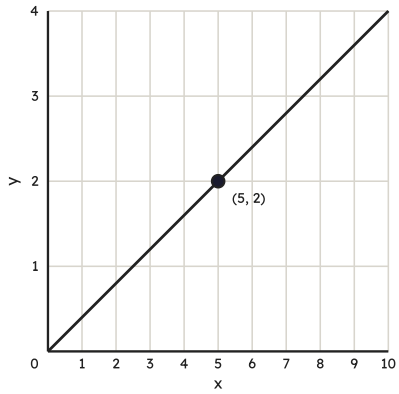
3. The graph shows the proportional relationship between minutes and pages printed.

What is the constant of proportionality, in pages per minute?



4. The graph shows a proportional relationship between x and y.

What is the constant of proportionality? Write your answer as a fraction in simplest form.



5. Each batch of paint is made with 2 cups of blue paint and 12 cups of white paint. The relationship between cups of blue paint and cups of white paint is proportional. Let x be the number of cups of white paint and let y be the number of cups of blue paint.

What is the constant of proportionality? What does it mean in this situation?

Score: ____ / 5 **My level: 1 2 3 4** **→ copy both to your progress strip**

Answer key — C-AR.4.2 · MA.7.AR.4.2 · Constant of proportionality

Unit 11: Representing Proportional Relationships. Regular = on-benchmark item; Rigor = the item that needs the benchmark's reasoning. Mastery on the weekly check = 4 of 5.

Weekly check (5 items)

1. Regular **A. 9.15**

The constant of proportionality is cups of water per minute: $64.05 \div 7 = 9.15$.

0.11 divides the other way, 64.05 is the total rather than the rate, and adding or subtracting the two amounts does not give a rate.

2. Regular **3.1**

The constant of proportionality is the fee divided by the length.

$62 \div 20 = 3.1$, and $68.2 \div 22 = 3.1$.

The constant of proportionality is 3.1 dollars per foot.

3. Regular **1.8 pages per minute**

Use a labeled point: $(5,\ 9)$.

The constant of proportionality is $y \div x = 9 \div 5 = 1.8$ pages per minute.

It is not the y-value of a labeled point, because no labeled point has $x = 1$.

4. Regular $\dfrac{2}{5}$

Use the labeled point: $y \div x = 2 \div 5 = \dfrac{2}{5}$.

5. Rigor **$k = \dfrac{1}{6}$; each cup of white paint is used with $\dfrac{1}{6}$ cup of blue paint. Score: 1 point for the constant (or an equivalent form of it) and its meaning; the constant alone earns 0.**

The constant of proportionality is $y \div x$: $2 \div 12 = \dfrac{1}{6}$.

So $y = \left(\dfrac{1}{6}\right) x$: each cup of white paint is used with $\dfrac{1}{6}$ cup of blue paint.

Dividing the other way, $12 \div 2 = 6$, would be the constant if x were the cups of blue paint; here x is the cups of white paint.

Re-check strip (3 items, C-AR.4.2-R)

1. Regular **0.9 cups of flour per batch**

Use a labeled point: $(5,\ 4.5)$.

The constant of proportionality is $y \div x = 4.5 \div 5 = 0.9$ cups of flour per batch.

It is not the y-value of a labeled point, because no labeled point has $x = 1$.

2. Regular **3.95**

The constant of proportionality is the fee divided by the length.

$63.2 \div 16 = 3.95$, and $79 \div 20 = 3.95$.

The constant of proportionality is 3.95 dollars per foot.

3. Regular **D. 3.15**

The constant of proportionality is liters of paint per wall: $12.6 \div 4 = 3.15$.

0.32 divides the other way, 12.60 is the total rather than the rate, and adding or subtracting the two amounts does not give a rate.

Comparison strip (2 items, C-AR.4.2-X) — used on a later week's re-check when this benchmark was *not* retaught

1. Regular $\dfrac{9}{7} = 1\dfrac{2}{7}$

Use the labeled point: $y \div x = 9 \div 7 = \dfrac{9}{7}$.

2. Regular $\dfrac{8}{11}$

The constant of proportionality is $y \div x$.

$8 \div 11 = \dfrac{8}{11}$, and every pair gives the same ratio.

Reteach plan — C-AR.4.2

MA.7.AR.4.2 · Constant of proportionality

Item that gave the most trouble (from both tally sheets):

What the mistake looks like:

Worked example for Tuesday (8–10 min):

Three practice items (one change at a time):

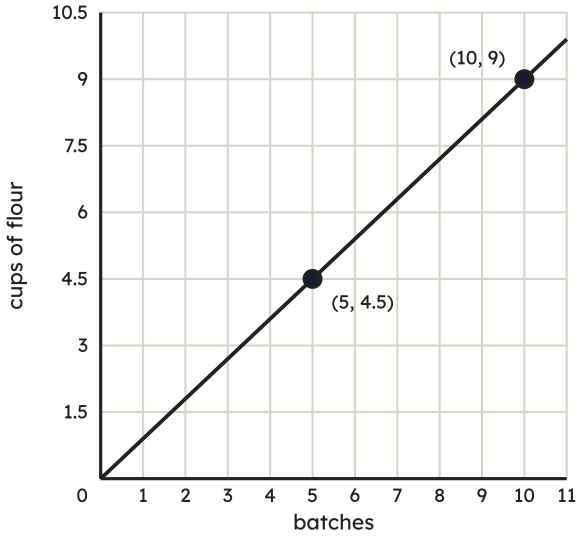
Decision line: Reteach on _____ Huddle: in person / paper Date: _____ Re-check given on _____

7th Grade Math · Re-check (3 items) C-AR.4.2-R

Benchmark: **MA.7.AR.4.2** · Constant of proportionality Student code: _____ Calculator: yes

1. The graph shows the proportional relationship between batches and cups of flour.

What is the constant of proportionality, in cups of flour per batch?



2. The daily fee for docking a boat at a marina is proportional to the length of the boat. The table shows the fee for four different boat lengths.

Boat length (feet)	Daily fee
16	$63.20
20	$79.00
22	$86.90
26	$102.70

What is the constant of proportionality, in dollars per foot?

3. In a proportional relationship, 12.6 liters of paint are needed to cover 4 walls.

What is the constant of proportionality, in liters of paint per wall?

A. 0.32

B. 16.60

C. 12.60

D. 3.15

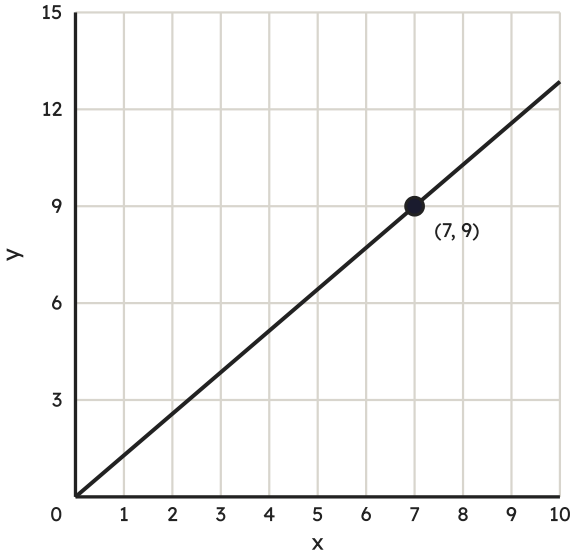
Score: ____ / 3 → copy to your progress strip

7th Grade Math · Comparison items (2) C-AR.4.2-X

Benchmark: **MA.7.AR.4.2** · Constant of proportionality Student code: _____ Calculator: yes

1. The graph shows a proportional relationship between x and y.

What is the constant of proportionality? Write your answer as a fraction in simplest form.



2. The table shows a proportional relationship between x and y.

x	y
11	8
22	16
55	40
66	48

What is the constant of proportionality? Write your answer as a fraction in simplest form.

Score: ____ / 2 → copy to your progress strip